

Orbit Guard AI

AI-Based Data Association for Short-Arc Space Object Tracking

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1. PROBLEM STATEMENT & MOTIVATION

The rapid proliferation of satellites and orbital debris has made Space Situational Awareness (SSA) a critical challenge. Modern phased-array radar and wide-field optical sensors generate massive volumes of "short-arc" observations -- brief, intermittent tracklets spanning seconds to minutes -- that provide severely limited kinematic information. Correlating these sparse measurements to known or unknown Resident Space Objects (RSOs) is the fundamental data association problem. Traditional methods such as Multi-Hypothesis Tracking (MHT) and Joint Probabilistic Data Association (JPDA) face combinatorial explosion in dense debris fields, while classical Extended Kalman Filters (EKF) fail under the highly non-linear dynamics and massive initial uncertainty inherent to short-arc orbit determination.

2. PROPOSED APPROACH

Orbit Guard AI is a full-stack, real-time space debris tracking system that fuses physics-informed estimation with AI-driven data association. The system ingests simulated multi-radar short-arc observations and maintains a persistent, live orbital catalog with conjunction assessment. The architecture comprises four tightly integrated modules:

- Ensemble Kalman Filter (EnKF):** A 200-particle non-linear state estimator with Admissible Region Sampling for initialization, anisotropic RTN-frame process noise, and adaptive noise scaling based on innovation statistics. Replaces linearized EKF/UKF with Monte Carlo covariance propagation.
- Dual Association Engine:** Switchable Global Nearest Neighbor (GNN) and JPDA associators. GNN uses Mahalanobis-gated assignment via the Hungarian algorithm; JPDA computes marginal association probabilities for ambiguous multi-target scenarios, with a chi-squared validation gate (3-DoF, 99% confidence).
- Physics-Informed Propagation:** High-fidelity RK45 numerical integration with J2 zonal harmonic, exponential atmospheric drag, and solar radiation pressure perturbations. Numba JIT-compiled guardrails enforce physical energy/altitude constraints on the particle ensemble.
- Real-Time Conjunction Assessment:** Pairwise miss-distance and collision probability (P_c) computation using Mahalanobis separation of combined covariance ellipsoids, with RED/YELLOW/SAFE risk classification.

3. SYSTEM ARCHITECTURE & IMPLEMENTATION

Backend: Python (NumPy, SciPy, Numba, Skyfield, FastAPI) implementing the tracker, scenario generator, TLE catalog correlator, and WebSocket streaming server. Frontend: React 18 + Three.js (React Three Fiber) providing a real-time 3D Earth visualization with live orbital trails, particle-cloud uncertainty rendering, telemetry dashboards, algorithm switching (GNN/JPDA), and conjunction alert overlays. The end-to-end pipeline processes frames at ~5s cadence with 3 simulated radar stations (ISTRAC Bangalore, SvalSat Norway, McMurdo Antarctica), streaming ECI-to-ECEF-transformed state vectors over WebSockets for globe-locked rendering.

4. KEY RESULTS

- Association Accuracy:** JPDA achieves >85% validated association rate on 5-10 object LEO scenarios with overlapping short-arc tracklets, outperforming GNN in dense, ambiguous configurations by 12-18%.
- Orbit Determination:** EnKF converges to <5 km position RMS within 10 measurement updates from angles-only initialization, with covariance traces reducing from ~50,000 sq.km to <500 sq.km.
- Conjunction Detection:** Successfully flags close approaches at <200 km threshold with calibrated P_c estimates; RED alerts ($P_c > 1e-4$) trigger within 3-hour lookahead windows.
- Real-Time Performance:** Full pipeline (propagation + association + conjunction) executes in <500ms per frame for 10 tracked objects with 200-particle ensembles, enabling live 3D visualization.

5. CONTRIBUTIONS & FUTURE WORK

This project demonstrates a complete, deployable SSA pipeline bridging AI-based multi-target data association with physics-constrained Bayesian estimation and interactive 3D visualization. Key contributions: (1) admissible-region particle initialization for short-arc scenarios, (2) adaptive RTN-frame process noise driven by innovation statistics, and (3) a live conjunction assessment engine with calibrated risk classification. Future work targets Graph Neural Network (GNN-DL) learned association metrics for >100 object scalability, integration of real TLE/CDM feeds from Space-Track.org, and physics-informed neural network (PINN) orbit propagation to replace numerical integration for faster-than-real-time catalog maintenance.

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